

Contextual Semantic Classification for Emergency Call Analysis Using Transformer-Based Models

1. The classification of emergency call transcripts into categories such as legitimate, prank, or scam represents a critical decision-making layer in the Rescue AI system. Unlike traditional rule-based approaches, which rely on predefined keyword matching, the use of transformer-based models enables a deeper understanding of contextual semantics, allowing the system to capture nuanced linguistic patterns that may indicate deception or urgency.

2. Theoretical Foundation

BERT (Bidirectional Encoder Representations from Transformers) is chosen due to its ability to model bidirectional context, which is particularly important in Urdu, where meaning is often dependent on surrounding words. The model processes input text as a sequence of tokens, generating contextual embeddings that capture both syntactic and semantic information.

The classification task is formulated as a supervised learning problem, where the model learns to map input sequences to discrete labels. The use of the [CLS] token as a summary representation allows the model to aggregate information across the entire sequence.

3. Dataset Design and Labeling Strategy

A significant challenge in training the classification model lies in constructing a high-quality labeled dataset. Given the lack of publicly available emergency call datasets in Urdu, a combination of synthetic data generation and manual annotation is employed. Care is taken to ensure that the dataset captures a wide range of linguistic variations, including informal speech, slang, and emotionally charged language.

Class imbalance is addressed through techniques such as oversampling and class weighting, ensuring that minority classes such as scam calls are adequately represented during training.

4. Training and Fine-Tuning

The model is fine-tuned from a pretrained multilingual BERT model, leveraging transfer learning to reduce the need for large amounts of labeled data. The training process involves minimizing cross-entropy loss, with regularization techniques such as dropout applied to prevent overfitting.

Hyperparameter tuning is conducted to identify optimal values for learning rate, batch size, and number of training epochs. Early stopping is employed to prevent degradation in validation performance.

5. Evaluation and Performance Analysis

Model performance is evaluated using metrics such as accuracy, precision, recall, and F1-score. Particular emphasis is placed on recall for legitimate emergency calls, as false negatives in this category could have severe consequences.

Error analysis reveals that misclassifications often occur in ambiguous cases where the linguistic cues are subtle or contradictory. This highlights the importance of incorporating additional features, such as acoustic signals, in future iterations.

6. Integration and Deployment

The BERT model is deployed as part of the backend inference pipeline, receiving input from the ASR module and returning classification results to the urgency detection system. The model is optimized for inference using techniques such as model pruning and quantization, ensuring that it meets real-time processing requirements.